

Mid-sem Examination (EP1118: Math for Physics)

Total marks: 30

Time: 2 hour

1. (a) Derive the arc length formula of a curve in two-dimensional space using parametrization. [3]
- (b) Derive the expression of line integral in various forms. [3]
- (c) Evaluate the line integral for the vector $F = y^2\hat{i} + 2x(y+1)\hat{j}$ between two points $a = (1, 1)$ and $b = (2, 2)$, joining by two curves C_1 and C_2 . [3]

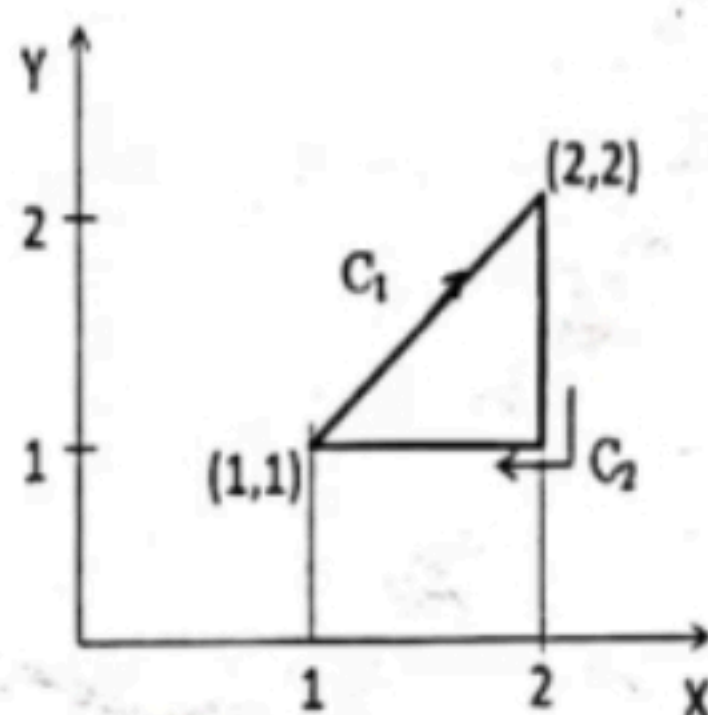


Figure 1. C_2 Curve

I. ANY 3 QUESTIONS

1. (a) Explain the gradient of a scalar field. Explain the connection between gradient of a scalar field and path independence of a line integral with required mathematical rigor. [3]

(b) Consider the vector field $\vec{F}(x, y, z) = (2x+y)\hat{i} + (x+z^3)\hat{j} + (3yz^2+1)\hat{k}$. Determine whether the $\vec{F}$ is conservative or not. If so, find its potential. [2]

(c) Derive the criteria for determination of conservative field by considering the vector field $\vec{F} = M(x, y, z)\hat{i} + N(x, y, z)\hat{j} + P(x, y, z)\hat{k}$ [2]

2. (a) Using Green's theorem in circulation form derive the Stokes theorem [4]

(b) Find the flux of $\vec{F} = xz\hat{i} + yz\hat{j} + z^2\hat{k}$ outward through the part of the sphere $x^2 + y^2 + z^2 = a^2$ lying in the first octant ($x, y, z, \geq 0$) [3].

3. (a) Derive the Green's theorem in flux-flow form. [4]

(b) Verify Stoke's theorem for the vector field $\vec{F} = (2x-y)\hat{i} - yz^2\hat{j} - y^2\hat{k}$, where S is the upper half of the sphere $x^2 + y^2 + z^2 = 1$ and C is the circular boundary on the $z = 0$ plane. [3]

4. (a) State and prove Divergence theorem. Explain its significance. [4]

(b) Verify the divergence theorem for the function $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the surface bounded by the cylinder $x^2 + y^2 = 4$ and $z = 0, z = 3$. [3]